DETAILED ANSWERS – MCA305T BIG DATA ANALYTICS

SECTION A

1. SVM Concepts in Data Classification
Support Vectors: These are critical data points closest to the decision boundary. For example, in a spam classification problem, emails closest to the boundary between spam and non-spam define the classifier.
Hyperplane: A decision boundary separating classes. In 2D it is a line; in 3D, a plane. Example: separating two types of fruits based on weight and color.
Kernel: Used when data is not linearly separable. Example: RBF kernel maps circular data into higher dimensions where it becomes separable.
Hard Margin: Used when data is perfectly separable (e.g., clearly distinct categories).
Soft Margin: Allows misclassification for noisy data. Example: real-world datasets like medical data often require soft margins.

2. MapReduce: Matrix-Vector Multiplication
Map Phase: Each mapper processes matrix elements and multiplies them with corresponding vector values. Example: For M[1,2] and v[2], output (1, M[1,2]*v[2]).
Reduce Phase: Reducer sums values for each row index. Example: Row 1 values summed to produce final result.
This is efficient for large-scale distributed systems like Hadoop.

3. Regression Modeling Methods
Linear Regression: Example – predicting house prices based on area.
Logistic Regression: Example – predicting whether a student passes (0 or 1).
Polynomial Regression: Example – modeling growth trends like population where data is curved.

5. Kernel Methods in SVM
Linear Kernel: Used for simple data (e.g., linearly separable text classification).
Polynomial Kernel: Useful in image processing where relationships are more complex.
Gaussian (RBF) Kernel: Example – face recognition or pattern detection with complex boundaries.

SECTION B

6. Cluster Analysis and Techniques (Improved Detailed Answer)
Cluster Analysis groups similar data points together. Example: grouping customers based on purchasing behavior.

Partitioning Methods (K-Means):
Divides data into K clusters. Example: segmenting customers into 3 groups based on spending.
Steps:
1. Choose K
2. Assign points to nearest centroid
3. Update centroid
4. Repeat until convergence

Hierarchical Methods:
Agglomerative (Bottom-Up): Start with individual points and merge clusters.
Divisive (Top-Down): Start with one cluster and split.
Example: Organizing documents into topic hierarchies.

Density-Based Methods (DBSCAN):
Clusters are formed based on dense regions.
Example: Detecting fraud transactions where normal data is dense and anomalies are sparse.
Advantages:
- Detects arbitrary shaped clusters

- Handles noise effectively

Types of Data in Cluster Analysis:
- Interval-scaled (e.g., age)
- Binary (yes/no)
- Nominal (categories)
- Ordinal (ranked data)

Overall, adding examples improves understanding and practical application of concepts.